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Received: 5 February 2026

Accepted: 24 March 2026

Published online: 01 April 2026

Cite this article as: Sun Y., Zhang F., Yan S. *et al.* DHGCMDA: a dual-view heterogeneous graph contrastive learning framework for miRNA-disease association type prediction. *BMC Bioinformatics* (2026). <https://doi.org/10.1186/s12859-026-06436-w>

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DHGCMDA: A Dual-view Heterogeneous Graph Contrastive Learning Framework for miRNA-Disease Association Type Prediction

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Abstract

Background: Accumulating evidence demonstrates that microRNA (miRNA) dysregulation drives the pathogenesis of diverse human diseases via intricate, context-dependent molecular mechanisms. Hence, prediction of miRNA-disease association types is a critical prerequisite for dissecting functional roles of miRNAs in disease initiation and progression. Although computational methods offer cost-effective, time-efficient alternatives to wet-lab experiments for miRNA-disease association type prediction, most of them are hampered by three key

limitations: excessive reliance on association-derived similarity metrics gives rise to quantification bias, traditional pairwise graph architectures inadequately capture high-order biological interactions, and existing representation learning strategies fail to generate consistent embeddings across heterogeneous views and modalities.

Results: To address these issues, this study presents DHGCMDA, a dual-view heterogeneous graph contrastive learning framework for miRNA-disease association type prediction. Specifically, dual-view hypergraphs are first constructed based on heterogeneous similarity data to avoid excessive reliance on association-derived similarity metrics. A hypergraph convolutional network is then employed to capture high-order topological relationships between miRNAs and diseases, with its convolution cooperatively integrated with contrastive learning, intra-modality for cross-view consistency and cross-modality for embedding space alignment, to enhance feature representation quality. Finally, an attention-guided adaptive view fusion strategy dynamically weights and integrates distinct view representations, and type-aware message passing via heterogeneous graph Transformer simultaneously enables prediction of association presence and functional types. 5-fold cross-validation on HMDD v2.0 and v3.2 datasets demonstrates that DHGCMDA outperforms several state-of-the-art methods. Furthermore, case studies on breast neoplasms and hepatocellular carcinoma reveal that most predicted association types are corroborated by published literature, thereby validating the efficacy of DHGCMDA in miRNA-disease association type prediction.

Conclusions: DHGCMDA exhibits robust discriminative power and generalization capability, providing a reliable computational alternative for miRNA-disease association type prediction. The source code is publicly available at <https://github.com/CDMBlab/DHGCMDA>.

Keywords: association type prediction, dual-view hypergraphs, heterogeneous graph transformer, contrastive learning, hypergraph convolutional network

1 Introduction

MicroRNAs (miRNAs) are a class of endogenous small non-coding RNAs [1] that act as pivotal regulators in diverse biological processes. Consequently, dysregulation of miRNAs is closely correlated with the development and progression of various human diseases [2]. Prediction of disease-associated miRNAs is regarded as a crucial step for elucidating pathogenic mechanisms and advancing targeted therapeutic strategies. Although conventional biological experiments remain valuable for validation purposes, these methods are typically characterized by time-consuming procedures,

intensive labor requirements, and limited throughput, which render large-scale systematic screening of miRNA-disease associations impractical. Rapid advances in high-throughput sequencing technologies have facilitated the development of multiple specialized databases dedicated to miRNA and disease studies. Notable among these resources, DIANA-TarBase [3], miRTarBase [4], and miRWalk [5] provide comprehensive miRNA-gene interaction data; MiRBase [6] supplies curated annotated miRNA sequences; MiREDiBase [7] focuses on miRNA editing site information; and the Comparative Toxicogenomics Database (CTD) [8] integrates a broad array of biological entities, including diseases, genes, phenotypes, and chemicals, together with the corresponding interaction networks among these entities. These databases have significantly enriched the current understanding of miRNA functionality and regulatory mechanisms, while also laying an essential foundation for the development of computational methods for the prediction of miRNA-disease associations. In parallel, the rapid evolution of representation learning paradigms, from traditional machine learning to deep graph models and multi-modal fusion, has further accelerated progress in biomedical data analysis [9].

Computational methods developed for miRNA-disease association prediction have undergone successive stages of evolution. Similarity-based methods [10, 11] were predicated on the assumption that functionally similar miRNAs tend to correlate with phenotypically similar diseases, and vice versa, yet such methods are inherently constrained by two critical limitations: an excessive reliance on similarity metrics and a fundamental inability to capture the intricate nonlinear relationships embedded in biological systems. Traditional machine learning methods, for instance, support vector machines and matrix factorization, achieved enhanced prediction accuracy; however, they typically processed miRNAs and diseases in an isolated manner and failed to explicitly model the intricate network architectures inherent to biological systems.

More recently, graph neural network (GNN) methods have demonstrated superior performance by explicitly modeling the topological relationships among biological entities through message passing mechanisms, and have been widely applied to miRNA-disease association prediction. Lu et al. [12] proposed a hypergraph learning method HGT-MDA that integrates an improved graph convolutional network (GCN)-Transformer encoder with a restart random walk strategy for predicting miRNA-disease associations. Liu et al. [13] developed GCNPCA, which combines GCN with attention mechanisms for topological feature extraction and fuses node attributes derived from principal component analysis (PCA) to support miRNA-disease association prediction. Ma et al. [14] introduced a self-feature-based graph autoencoder SFGAE designed to address excessive smoothing in deep graph models while enabling miRNA-disease association prediction. Li et al. [15] proposed an ensemble model GCSENet that integrates GCN, convolutional neural networks, and squeeze-and-excitation networks to achieve feature aggregation from heterogeneous networks for miRNA-disease association prediction. Ning et al. [16] developed HHAWMD, which employs hierarchical hypergraph learning in association-weighted heterogeneous networks to capture high-order dependencies among biological entities for miRNA-disease association identification. These methods, despite their demonstrated efficacy in binary association prediction, predominantly frame miRNA-disease relationships as simple presence-absence links, thereby overlooking the diverse biological mechanisms that underpin such associations. Accumulating evidence indicates that miRNA dysregulation drives disease pathogenesis via diverse underlying molecular mechanisms [17]. For instance, miR-146a exhibits dual roles in gastric cancer, promoting cancer progression via direct targeting of the SMAD4 gene to modulate cell proliferation and apoptosis [18], while its ectopic expression exerts tumor-suppressive effects by suppressing gastric cancer cell motility and reducing both EGFR and IRAK1 expression [19]. This mechanistic heterogeneity highlights that accurate prediction of miRNA-disease association types is critical for a deeper

understanding of the molecular mechanisms driving disease initiation and progression. Therefore, the prediction of association types represents a cutting-edge advancement in research on miRNA-disease association prediction.

In fact, several methods have been proposed for miRNA-disease association type prediction. Chen et al. proposed RBMMMDA [20], which constructs a restricted Boltzmann model for predicting association types between miRNAs and diseases. However, it only utilizes known miRNA-disease association information, without incorporating miRNA and disease similarity data to assist prediction. Zhang et al. developed NLPMMDA [21], which constructs a heterogeneous network and applies label propagation to predict miRNA-disease association types. Yet, this method treats the prediction of each type as an independent task, ignoring correlations among different types. Huang et al. proposed TDRC [22], which models miRNA-disease association type data as a tensor, incorporates miRNA and disease similarity matrices as constraint terms, and predicts novel miRNA-disease association types via canonical polyadic decomposition. Ouyang et al. developed two tensor factorization methods, WeightTDAIGN [23] and SPLDHyperAWNTF [24], to identify potential miRNA-disease association types. The former incorporates graph Laplacian regularization, L2,1-norm constraints on projection matrices, and multi-similarity network integration to enhance predictive capability. The latter combines self-learning with nonnegative tensor factorization, adopts hypergraph regularization, and introduces an adaptive weight tensor to mitigate the impact of false-negative samples. Yu et al. proposed TFLP [25], which obtains low-rank association tensors based on tensor robust PCA and employs label propagation to predict potential miRNA-disease association types. Nevertheless, false negatives, high time complexity, and data sparsity represent common challenges faced by tensor factorization based miRNA-disease association type prediction methods. Research on the prediction of miRNA-disease association types based on GNNs has just emerged, but encouraging results have already been achieved.

Yan et al. proposed PDMDA [26], which extracts miRNA feature representations via fully connected networks, leverages GNNs to capture disease feature representations, and designs a multi-layer structured classifier to predict miRNA-disease association types. Zhang et al. regarded miRNA-disease associations as signed bipartite networks, and proposed a signed GNN method SGNNMD [27] for predicting the upregulated/-downregulated expression types of miRNAs in diseases. SGNNMD extracts subgraphs surrounding known miRNA-disease associations, learns the structural features of these subgraphs through node feature extraction and GNNs, and integrates such structural features with biological features to construct a prediction model. Wang et al. developed NMCMDA [28], which applies relational GCNs to miRNA-disease heterogeneous networks for learning the latent space embeddings of miRNAs and diseases respectively, and utilizes a neural multi-relational decoder to calculate predictive scores for miRNA-disease association types. Yu et al. designed mDLinker [29], which adopts the node2vec graph representation learning to extract node features from disease and miRNA similarity networks, captures topological information and multi-type nonlinear relationships between miRNAs and diseases based on the GATNE heterogeneous graph multi-relational model, and trains a random forest classifier to predict miRNA-disease association types. Liu et al. [30] proposed deepMDpred, which combines GCN with a multichannel attention mechanism to extract disease features from heterogeneous similarity and gene interaction networks, and fuses the learned miRNA sequence-structural features with disease representations via a nonlinear fully connected layer for multi-type miRNA-disease association prediction. However, most GNN-based methods are constrained by three limitations. First, calculating miRNA similarity based on known miRNA-disease associations leads to excessive reliance on existing association data, which not only creates circular dependencies that render the method ineffective for novel miRNAs lacking prior disease annotations but also induces quantification bias

accordingly. Second, traditional pairwise graph structures can only model binary relationships between nodes, failing to capture either the complex multi-way dependencies prevalent in biological systems or the diverse high-order biological interactions. Third, existing representation learning strategies fail to generate consistent embeddings across heterogeneous views and modalities, thereby impairing their ability to integrate corresponding biological information. Although Zhu et al. [31] proposed SMCLMDA, which introduces self-supervised multi-channel contrastive learning to construct contrastive objectives across different data channels for learning noise-robust miRNA-disease representations, cross-view and cross-modality consistency in the context of multi-type association prediction remains insufficiently explored.

To address these challenges, this study presents a dual-view heterogeneous graph contrastive learning framework, abbreviated as DHGCMMDA, for the prediction of miRNA-disease association types. Accordingly, the main contributions of DHGCMMDA are fourfold:

- Dual-view hypergraphs are constructed based on heterogeneous similarity data, thus avoiding excessive reliance on association-derived similarity metrics and enabling robust prediction for novel miRNAs without prior disease annotations.
- A hypergraph convolutional network (HGCN) is employed to capture high-order topological relationships between miRNAs and diseases, beyond traditional pairwise connections, by leveraging hyperedges to model multi-way dependencies in biological systems.
- Hypergraph convolution is integrated with contrastive learning to enhance representation learning, where intra-modality contrastive learning enforces consistency across complementary views of the same entity and cross-modality contrastive learning aligns embedding spaces of miRNAs and diseases.

- An attention-guided adaptive view fusion strategy is employed to dynamically weight and integrate distinct view representations, while type-aware message passing via a heterogeneous graph Transformer (HGT) enables simultaneous prediction of miRNA-disease association presence and functional types.

2 Methods

2.1 Overview of DHGCMDBA

The overview of DHGCMDBA is illustrated in Fig. 1. First, dual-view hypergraphs corresponding to miRNAs and diseases are constructed based on heterogeneous similarity data. Second, a HGCN is employed to capture high-order topological relationships between miRNAs and diseases, with hypergraph convolution integrated with contrastive learning, where intra-modality contrastive learning ensures cross-view consistency and cross-modality contrastive learning aligns embedding spaces, to enhance feature representation quality. Finally, an attention-guided adaptive view fusion strategy is utilized to dynamically weight and integrate distinct view representations, while type-aware message passing implemented via a HGT enables simultaneous prediction of association presence and functional types.

2.2 Problem Formulation and Data Representation

The prediction of miRNA-disease associations is formulated as a link prediction problem in a heterogeneous biological network. Let $\mathcal{M} = \{m_1, m_2, \dots, m_{n_m}\}$ denote the set of n_m miRNAs, and $\mathcal{D} = \{d_1, d_2, \dots, d_{n_d}\}$ denote the set of n_d diseases. Their association relationships are encoded in a matrix $\mathbf{A} \in \mathbb{R}^{n_m \times n_d}$, where each entry A_{ij} quantifies the association between miRNA m_i and disease d_j . For binary association prediction, A_{ij} takes a binary value of either 1 or 0, where 1 indicates a confirmed association between m_i and d_j . It should be noted that $A_{ij} = 0$ does not imply the absence of an association between m_i and d_j , but rather indicates that such a

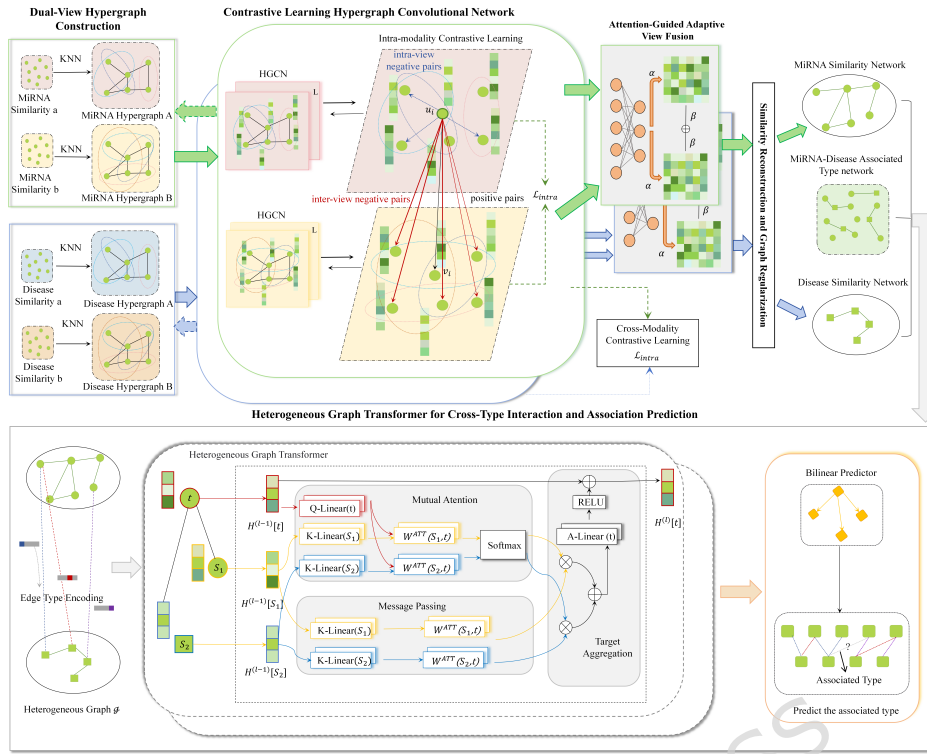


Fig. 1 Overview of DHGCMMA.

relationship has not yet been confirmed. Similarly, for association type prediction, $A_{ij} \in \{0, 1, 2, \dots, C\}$, where C denotes the total number of distinct association types, and the value 0 retains the same connotation of unconfirmed association as defined in the binary prediction task.

To obtain robust and comprehensive feature representations, dual complementary views are constructed for miRNAs and diseases based on heterogeneous similarity data. For miRNAs, $\mathbf{S}_m^{seq} \in \mathbb{R}^{n_m \times n_m}$ represents sequence-based similarity and $\mathbf{S}_m^{func} \in \mathbb{R}^{n_m \times n_m}$ represents functional similarity. For diseases, $\mathbf{S}_d^{gene} \in \mathbb{R}^{n_d \times n_d}$ represents gene-based similarity and $\mathbf{S}_d^{sem} \in \mathbb{R}^{n_d \times n_d}$ represents semantic similarity.

Besides, intra-modality similarity information and cross-modality association information are integrated to construct augmented feature matrices for each view via concatenation of similarity matrices and the association matrix,

$$\mathbf{X}_m^{(v)} = [\mathbf{S}_m^{(v)}, \mathbf{A}] \in \mathbb{R}^{n_m \times (n_m + n_d)} \quad (1)$$

$$\mathbf{X}_d^{(v)} = [\mathbf{S}_d^{(v)}, \mathbf{A}^T] \in \mathbb{R}^{n_d \times (n_d + n_m)} \quad (2)$$

where v denotes the view type. This augmentation ensures that feature representation of each node encodes both intrinsic biological properties through intra-modality similarity and functional interaction patterns through cross-modality associations.

2.3 Dual-View Hypergraph Construction

Traditional pairwise graph structures exhibit inherent limitations in capturing the complex multi-way dependencies prevalent in biological systems. By contrast, hypergraphs enable more expressive representations via hyperedges that connect arbitrary node subsets, thus facilitating the modeling of higher-order relationships. For each of these four augmented views, a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E}, \mathcal{W})$ is constructed, where $\mathcal{V}$ denotes the node set, $\mathcal{E}$ represents the hyperedge set, and $\mathcal{W}$ assigns importance weights to individual hyperedges.

Specifically, a K -nearest neighbors (KNN) strategy is adopted to generate locality-preserving hyperedges from augmented feature matrices. For each node v_i , its K nearest neighbors are identified based on cosine similarity in the augmented feature space, and a hyperedge $e_i = \{v_i\} \cup KNN(v_i, K)$ is constructed accordingly. The hypergraph structure is formally encoded using an incidence matrix $\mathbf{H} \in \{0, 1\}^{n \times |\mathcal{E}|}$,

$$H_{ij} = \begin{cases} 1, & \text{if node } v_i \in \text{hyperedge } e_j \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where n denotes the number of nodes and $|\mathcal{E}|$ denotes the number of hyperedges.

Efficient information propagation across the hypergraph is enabled by computing a normalized hypergraph Laplacian matrix $\mathbf{G}$, which functions as the propagation operator. The normalized Laplacian is formulated as,

$$\mathbf{G} = \mathbf{D}_v^{-1/2} \mathbf{H} \mathbf{W} \mathbf{D}_e^{-1} \mathbf{H}^T \mathbf{D}_v^{-1/2} \quad (4)$$

where $\mathbf{D}_v$ denotes the node degree matrix, $\mathbf{D}_e$ denotes the hyperedge degree matrix, and $\mathbf{W}$ denotes the diagonal hyperedge weight matrix (by default $\mathbf{W} = \mathbf{I}$ for unweighted hypergraph). Intuitively, the information propagation defined by this normalized Laplacian can be understood as a two-phase process: in the first phase, node features are aggregated into their associated hyperedges (via $\mathbf{H}^T$), with $\mathbf{D}_e^{-1}$ normalizing by hyperedge cardinality so that large hyperedges do not dominate the aggregation; in the second phase, the aggregated hyperedge representations are redistributed back to constituent nodes (via $\mathbf{H}$), with $\mathbf{D}_v^{-1/2}$ applied symmetrically on both sides to ensure that high-degree nodes neither dominate the message passing nor receive disproportionately amplified signals. Through this two-phase mechanism, the formulation effectively generalizes the symmetric normalization of standard graph Laplacians to the hypergraph setting, thereby preserving inherent multi-way connectivity while ensuring numerically stable feature propagation.

2.4 Contrastive Learning Hypergraph Convolutional Network

Contrastive learning has emerged as a powerful paradigm for enhancing graph embedding quality in supervised learning scenarios. By optimizing contrastive objectives designed to discriminate between positive and negative sample pairs, highly discriminative and robust graph feature representations are learned. Hence, a hypergraph contrastive learning method is proposed herein, which capitalizes on both intra-modality consistency across complementary views and cross-modality alignment between miRNAs and diseases.

Based on each constructed hypergraph structure, HGCN with spectral convolution is employed to encode intrinsic high-order relations. A hyperedge convolution layer of HGCN is formulated as,

$$\mathbf{X}^{(l+1)} = \sigma \left(\mathbf{G}\mathbf{X}^{(l)}\Theta^{(l)} \right) \quad (5)$$

where $\mathbf{X}^{(l)}$ is the aggregated information of hypergraph at layer l , with the initial augmented feature matrix $\mathbf{X}^{(0)} = \mathbf{X}$, $\Theta^{(l)}$ is the learnable filter matrix of the l -th layer, and $\sigma(\cdot)$ denotes the nonlinear activation function.

The contrastive learning HGCN architecture consists of two parallel HGCN branches. These branches are tasked with processing augmented features and hypergraph structures from distinct complementary views, thereby yielding view-specific node representations $\mathbf{Z}_1$ and $\mathbf{Z}_2$. Furthermore, a projection head is incorporated to project these embeddings into a normalized feature space, which facilitates more effective contrastive optimization.

2.4.1 Intra-modality Contrastive Learning

For any node v_i , its embedding generated from one view, denoted as $\mathbf{u}_i$, is designated as the anchor. The counterpart embedding of node v_i derived from the other view is represented as $\mathbf{v}_i$. Specifically, the embedding pair $\mathbf{u}_i$ and $\mathbf{v}_i$ of the same node across two views constitutes a positive sample pair. In contrast, $\mathbf{u}_i$ and $\mathbf{v}_i$ are treated as negative samples when paired with all other embeddings $\mathbf{u}_k$ and $\mathbf{v}_k$ ($k \neq i$) from either of the two views. Notably, the embedding $\mathbf{u}_k$ of other nodes within the same view as the anchor $\mathbf{u}_i$ forms intra-view negative pairs with $\mathbf{u}_i$, whereas the embedding $\mathbf{v}_k$ of other nodes from the opposite view forms inter-view negative pairs with $\mathbf{u}_i$.

The pairwise training objective for each positive pair $(\mathbf{u}_i, \mathbf{v}_i)$ is formally defined as,

$$\ell_{CL}(\mathbf{u}_i, \mathbf{v}_i) = -\log \frac{\exp(\theta(\mathbf{u}_i, \mathbf{v}_i)/t)}{\Phi_i} \quad (6)$$

where the normalization term Φ_i is written as,

$$\Phi_i = \exp(\theta(\mathbf{u}_i, \mathbf{v}_i)/t) + \sum_{k \neq i} \exp(\theta(\mathbf{u}_i, \mathbf{u}_k)/t) + \sum_{k \neq i} \exp(\theta(\mathbf{u}_i, \mathbf{v}_k)/t) \quad (7)$$

where t denotes a temperature parameter that regulates the concentration of similarity distributions. The similarity function $\theta(\mathbf{u}, \mathbf{v})$ is formulated as,

$$\theta(\mathbf{u}, \mathbf{v}) = s(g(\mathbf{u}), g(\mathbf{v})) \quad (8)$$

where $s(\cdot, \cdot)$ represents the cosine similarity and $g(\cdot)$ denotes a nonlinear projection head, serving to enhance the discriminative power of the similarity evaluation. To ensure symmetric contrastive learning across both views, the final bidirectional intra-modality contrastive loss is constructed as,

$$\mathcal{L}_{intra} = \alpha \sum_{i=1}^n \ell_{CL}(\mathbf{u}_i, \mathbf{v}_i) + (1 - \alpha) \sum_{i=1}^n \ell_{CL}(\mathbf{v}_i, \mathbf{u}_i) \quad (9)$$

where n denotes the total number of nodes within a single modality (miRNAs or diseases), and α is assigned to a default value of 0.5 to enforce balanced contributions from the two view directions.

This intra-modality contrastive learning operates on single-modal data, by leveraging complementary view pairs to guide the model to filter out view-specific noise and retain the inherent biological identities of individual entities. The resulting view-invariant representations are robust and discriminative, laying a solid foundation for subsequent cross-modality alignment between miRNAs and diseases, thereby facilitating the prediction of their associations.

2.4.2 Cross-Modality Contrastive Learning

Beyond enforcing consistency within each modality, cross-modality contrastive learning is introduced to explicitly model associations between miRNAs and diseases via aligning their respective embedding spaces.

Based on the fused miRNA and disease representations, denoted as $\mathbf{Z}_M \in \mathbb{R}^{n_m \times d}$ and $\mathbf{Z}_D \in \mathbb{R}^{n_d \times d}$, which are obtained by the attention-guided adaptive view fusion strategy, the detailed description of which can be found in Section 2.5, the cross-modal similarity is defined as,

$$\mathbf{S}_{cross} = \frac{\hat{\mathbf{Z}}_M \hat{\mathbf{Z}}_D^T}{t} \quad (10)$$

where $\hat{\mathbf{Z}}_M$ and $\hat{\mathbf{Z}}_D$ are normalized embeddings, formulated as,

$$\hat{\mathbf{Z}}_M = \frac{\mathbf{Z}_M}{\|\mathbf{Z}_M\|_2} \quad (11)$$

$$\hat{\mathbf{Z}}_D = \frac{\mathbf{Z}_D}{\|\mathbf{Z}_D\|_2} \quad (12)$$

The cross-modality contrastive loss is formulated as a synergistic combination of the InfoNCE [32] loss and a margin-based ranking loss. For each miRNA i and its corresponding associated diseases (positive samples), the InfoNCE loss is defined as,

$$\mathcal{L}_{inter}^{InfoNCE} = - \sum_i \sum_{j \in \mathcal{P}_i} \log \frac{\exp(S_{cross,ij})}{\exp(S_{cross,ij}) + \sum_{k \in \mathcal{N}_i} \exp(S_{cross,ik})} \quad (13)$$

where $\mathcal{P}_i = \{j | A_{ij} = 1\}$ denotes the set of diseases associated with miRNA i , and $\mathcal{N}_i$ denotes the set of sampled negative pairs for miRNA i . To further improve the separability between positive and negative sample representations and mitigate the issue of sample imbalance, the margin-based ranking loss is incorporated,

$$\mathcal{L}_{inter}^{Margin} = \max(0, m - (\bar{S}_{\mathcal{P}} - \bar{S}_{\mathcal{N}})) \quad (14)$$

where $\bar{S}_{\mathcal{P}} = \text{mean}(\mathbf{S}_{cross}[\mathcal{P}])$, $\bar{S}_{\mathcal{N}} = \text{mean}(\mathbf{S}_{cross}[\mathcal{N}])$, and m is a predefined margin hyperparameter. Hence, the cross-modality contrastive loss can be written as,

$$\mathcal{L}_{inter} = \mathcal{L}_{inter}^{InfoNCE} + \lambda \cdot \mathcal{L}_{inter}^{Margin} \quad (15)$$

where λ is a weighting coefficient that balances contributions of these two loss components.

2.5 Attention-Guided Adaptive View Fusion

Different similarity views exhibit heterogeneous contributions to association type prediction, with their relative importance varying across distinct biological contexts. To address this issue, an attention-guided adaptive view fusion is applied to dynamically weight the contribution of each view.

For each modality, given two view-specific embeddings $\mathbf{Z}_1$ and $\mathbf{Z}_2$ obtained from the contrastive learning HGCN, an attention mechanism is employed to adaptively integrate these embeddings. Specifically, for the v -th view where $v \in \{1, 2\}$, a global representation is first derived via global average pooling, and the attention weight α_v is subsequently computed using a two-layer fully connected network. This process is formulated as,

$$\alpha_v = \sigma(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \text{GAP}(\mathbf{Z}_v))) \quad (16)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable weight matrices corresponding to the two linear layers, $\text{GAP}(\cdot)$ denotes the global average pooling function, $\text{ReLU}(\cdot)$ represents the ReLU activation function, and σ denotes the sigmoid activation function used to constrain $\alpha_v \in [0, 1]$.

The attention weight α_v is applied in an element-wise manner to modulate the embeddings of each view, defined as,

$$\tilde{\mathbf{Z}}_v = \text{ReLU}(\alpha_v \mathbf{Z}_v) \quad (17)$$

and the final fused representation is obtained through a learnable weighted combination, formulated as,

$$\mathbf{Z}_{fused} = \sum_{v=1}^2 \beta_v \tilde{\mathbf{Z}}_v \quad (18)$$

where β refers to the set of learnable fusion coefficients optimized in an end-to-end fashion. This two-stage gating mechanism offers enhanced flexibility: attention weights capture instance-specific view importance, while fusion coefficients encode global view reliability. Identical fusion procedures are implemented independently for miRNA and disease views, yielding the miRNA fused representation $\mathbf{Z}_M \in \mathbb{R}^{n_m \times d}$ and the disease fused representation $\mathbf{Z}_D \in \mathbb{R}^{n_d \times d}$.

2.6 Similarity Reconstruction and Graph Regularization

To ensure that learned embeddings retain topological structures inherent to the original similarity networks, a similarity reconstruction module is designed, which projects fused node representations back to the similarity space, enforcing a one-to-one correspondence between geometrically proximate nodes in the embedding space and functionally similar biological entities.

The reconstruction process employs a two-layer feedforward network, formulated as,

$$\mathbf{Z}_{proj} = \mathbf{W}_{proj}^2 \text{ReLU}(\mathbf{W}_{proj}^1 \mathbf{Z}_{fused}) \quad (19)$$

where $\mathbf{W}_{proj}^1 \in \mathbb{R}^{d \times (d/2)}$ and $\mathbf{W}_{proj}^2 \in \mathbb{R}^{(d/2) \times d}$ denote the learnable weight matrices of two linear layers in the feedforward network. Pairwise similarities are then computed as,

$$\hat{\mathbf{S}} = \mathbf{Z}_{proj} \mathbf{Z}_{proj}^T \quad (20)$$

The reconstruction loss is derived by comparing the reconstructed similarity matrices with the original counterparts using the Frobenius norm, given by,

$$\mathcal{L}_{recon} = \|\mathbf{S}_M - \hat{\mathbf{S}}_M\|_F^2 + \|\mathbf{S}_D - \hat{\mathbf{S}}_D\|_F^2 \quad (21)$$

This reconstruction objective serves multiple purposes: it acts as a regularizer to mitigate overfitting, provides a quantitative means to validate the quality of learned representations, and critically, the reconstructed similarity matrices are leveraged to dynamically update the heterogeneous graph structure during training, a strategy that implements graph regularization, enabling DHGCMDA to refine its structural assumptions as the quality of representations improves while imposing structural consistency constraints between the embedding space and graph space.

2.7 Heterogeneous Graph Transformer

After dual-view feature extraction and fusion, a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ is constructed to explicitly model relationships among miRNAs, diseases, and their associations. The node set $\mathcal{V} = \mathcal{M} \cup \mathcal{D}$ consists of miRNA nodes and disease nodes, with each node assigned the corresponding fused embeddings $\mathbf{Z}_M$ and $\mathbf{Z}_D$, respectively. The edge set $\mathcal{E}$ encompasses four distinct types, namely, miRNA-miRNA similarity edges weighted by $\hat{\mathbf{S}}_M$, disease-disease similarity edges weighted by $\hat{\mathbf{S}}_D$, miRNA-disease association edges, and disease-miRNA association edges.

To enable type-aware message passing across this heterogeneous structure, a HGT architecture is employed. Unlike homogeneous GNNs, HGT incorporates type-specific parameter settings to capture distinct semantic roles of different biological entities. Before feeding into HGT layers, type-specific linear projections are applied as,

$$\mathbf{X}_{miRNA} = \mathbf{W}_{miRNA} \mathbf{Z}_M \quad (22)$$

$$\mathbf{X}_{disease} = \mathbf{W}_{disease} \mathbf{Z}_D \quad (23)$$

where $\mathbf{W}_{miRNA}$ and $\mathbf{W}_{disease}$ represent type-specific learnable projection matrices, $\mathbf{X}_{miRNA}$ and $\mathbf{X}_{disease}$ are the projected feature representations tailored for subsequent HGT processing. For each node i with type $\tau(i)$, the HGT layer computes type-aware query, key, and value representations,

$$\mathbf{Q}_i^{\tau(i)} = \mathbf{W}_Q^{\tau(i)} \mathbf{x}_i \quad (24)$$

$$\mathbf{K}_i^{\tau(i)} = \mathbf{W}_K^{\tau(i)} \mathbf{x}_i \quad (25)$$

$$\mathbf{V}_i^{\tau(i)} = \mathbf{W}_V^{\tau(i)} \mathbf{x}_i \quad (26)$$

where $\mathbf{W}_Q^{\tau(i)}$, $\mathbf{W}_K^{\tau(i)}$, $\mathbf{W}_V^{\tau(i)}$ represent type-specific projection matrices, and $\mathbf{x}_i$ denotes the projected feature representation of node i . These type-aware query, key, and value representations lay the foundation for type-specific message passing, where messages between nodes are adaptively weighted based on their entity types and edge relations.

The attention coefficient between source node j and target node i connected by edge type r is computed via multi-head attention with edge-type-specific transformations,

$$\alpha_{ij}^r = \text{softmax}_{j \in \mathcal{T}(i)} \left(\frac{(\mathbf{W}_{att}^r \mathbf{Q}_i^{\tau(i)})^T (\mathbf{W}_{msg}^r \mathbf{K}_j^{\tau(j)})}{\sqrt{d_k}} \right) \quad (27)$$

where $\mathbf{W}_{att}^r$ and $\mathbf{W}_{msg}^r$ denote edge-type-specific transformation matrices for relation type r , $\mathcal{T}(i)$ denotes the neighborhood set of target node i , and d_k refers to the dimension of the query/key vectors in the attention mechanism. The scaled dot-product attention with $1/\sqrt{d_k}$ scaling mitigates the gradient vanishing issue during model training.

Node representations are updated via aggregation of attention-weighted value projections,

$$\mathbf{x}'_i = \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{T}_i^r} \alpha_{ij}^r (\mathbf{W}_{msg}^r \mathbf{V}_j^{\tau(j)}) \quad (28)$$

where $\mathcal{R}$ denotes the set of all edge types, $\mathcal{T}_i^r$ represents the subset of neighbors of node i connected through relation type r . Residual connections, dropout, and layer normalization are further incorporated to enhance model stability and alleviate overfitting,

$$\mathbf{x}_i^{(l+1)} = \text{LayerNorm}(\mathbf{x}_i^{(l)} + \text{Dropout}(\mathbf{x}'_i)) \quad (29)$$

where $\mathbf{x}_i^{(l)}$ denotes the representation of node i at the l -th HGT layer.

With two stacked HGT layers and four attention heads per layer, through multi-layer type-aware message passing, DHGCMDBA progressively refines node embeddings by integrating multi-hop heterogeneous neighborhood information across distinct biological entity types. The refined final embeddings of miRNA and disease nodes are subsequently fed into a bilinear predictor, which enables simultaneous prediction of two critical tasks: the existence of miRNA-disease associations and the corresponding functional types underlying these associations.

2.8 Dynamic Hypergraph Update Strategy

To enable co-evolution of graph structure and learned representations, a dynamic hypergraph update strategy is introduced. Initially, the graph is constructed using raw similarity matrices retrieved from biological databases, which inevitably contain measurement noise. As training proceeds and node embeddings are progressively refined, the reconstructed similarity matrices $\hat{\mathbf{S}}_M$ and $\hat{\mathbf{S}}_D$ can better characterize the true underlying biological relationships.

Every five training epochs, the within-type similarity edges in the heterogeneous graph $\mathcal{G}$ are regenerated. The reconstructed similarity matrices are subjected to thresholding to construct binary edge sets,

$$\mathcal{E}_M = \{(m_i, m_j) | \hat{S}_{M,ij} > \theta \wedge i \neq j\} \quad (30)$$

$$\mathcal{E}_D = \{(d_i, d_j) | \hat{S}_{D,ij} > \theta \wedge i \neq j\} \quad (31)$$

where θ denotes the similarity threshold, set to 0.5. The reconstructed similarity scores are normalized to $[0, 1]$, and 0.5 serves as a natural midpoint that partitions node pairs into higher- and lower-similarity groups, balancing noise filtering and structural connectivity preservation, consistent with common thresholding practices in biological network construction. Moreover, since the dynamic update mechanism regenerates edges every five epochs using progressively refined similarity matrices, the resulting graph topology evolves as embedding quality improves, reducing the dependence on the precise choice of this threshold.

The updated edge sets replace the original similarity edges, whereas the association edges are kept unchanged to preserve ground-truth supervisory signals critical for model training. Additionally, hypergraphs $\mathcal{H}$ are reconstructed using the KNN strategy based on the updated reconstructed similarity matrices.

2.9 Unified Training Objective

The unified training objective integrates multiple loss components to jointly optimize representation learning, contrastive alignment, similarity preservation, and miRNA-disease association type prediction,

$$\mathcal{L}_{total} = \mathcal{L}_{type} + \lambda_1 \mathcal{L}_{intra} + \lambda_2 \mathcal{L}_{inter} + \lambda_3 \mathcal{L}_{recon} \quad (32)$$

where $\lambda_1, \lambda_2, \lambda_3$ are trade-off hyperparameters that balance contributions of different loss components, and $\mathcal{L}_{type}$ denotes the weighted cross-entropy loss for association type prediction, which incorporates effective number weighting to handle imbalanced type distribution,

$$\mathcal{L}_{type} = - \sum_{(i,j) \in \mathcal{F}} \sum_{k=1}^C w_k y_{ij}^k \log(\hat{p}_{ij}^k) \quad (33)$$

where i and j denote the miRNA node and disease node respectively, $\mathcal{F}$ represents the set of all miRNA-disease association pairs, $y_{ij}^k \in \{0, 1\}$ is the binary ground-truth label (1 indicates that the i -th miRNA and j -th disease are associated with the k -th type, and 0 otherwise), $\hat{p}_{ij}^k$ is the predicted probability of the i -th miRNA and j -th disease having the k -th association type, w_k are class weights computed via,

$$w_k = \frac{1 - \beta}{1 - \beta^{n_k}} \quad (34)$$

where n_k denotes the number of samples belonging to the k -th association type, and β is selected to be sufficiently close to 1 for effective minority class weighting.

DHGCMDA is optimized using the Adam optimizer, coupled with suitable learning rate scheduling and early stopping criteria determined by validation set performance. Furthermore, dropout regularization is incorporated during the training phase to enhance the generalization capability and mitigate overfitting.

3 Results and Discussion

3.1 Dataset

The Human MicroRNA Disease Database (HMDD) serves as a widely adopted benchmark resource for miRNA-disease association prediction [17]. HMDD v2.0 has been extensively leveraged as a foundational data source in computational studies dedicated to this task, in which curated associations are classified into four distinct types based on evidence from genetics, epigenetics, circulation, and miRNA-target interactions. Building on this classification framework, HMDD v3.2 delivers substantial expansions in dataset coverage, with more miRNAs, diseases, and validated associations included.

Table 1 Statistics of benchmark datasets.

Datasets	miRNAs	Diseases	Associations	Types
HMDD v2.0	495	383	1,679	4
HMDD v3.2	411	271	11,748	5

Table 2 Distribution of association types in benchmark datasets.

Datasets	Genetics	Epigenetics	Circulation	Target	Tissue
HMDD v2.0	681 (40.56%)	199 (11.85%)	443 (26.38%)	356 (21.20%)	–
HMDD v3.2	1,155 (9.83%)	403 (3.43%)	2,293 (19.52%)	3,997 (34.02%)	3,900 (33.20%)

This updated version further introduces an additional association type (tissue) and adopts more refined and accurate criteria for evidence classification.

In this study, an identical standardized preprocessing pipeline is implemented for both HMDD v2.0 and HMDD v3.2 to ensure data density and reliability. Specifically, miRNAs and diseases with fewer than two experimentally verified associations are excluded, and the remaining data subsets are utilized for experiments, with detailed statistics summarized in Table 1 and Table 2. As shown in Table 2, the distribution of association types exhibits notable imbalance in both datasets. In HMDD v2.0, genetics accounts for the largest proportion (40.56%), whereas epigenetics represents only 11.85%. In HMDD v3.2, the imbalance is even more pronounced, with epigenetics constituting merely 3.43% of all associations compared to 34.02% for target type. This substantial class imbalance motivates the adoption of class-weighted cross-entropy loss (Eq. 34) to prevent the model from being biased toward majority types.

3.2 Evaluation settings

Two types of 5-fold cross-validation were employed to comprehensively evaluate method performance in predicting miRNA-disease association types.

- **CV_{triplet}**: All known miRNA-disease-type triplets were randomly partitioned into 5 equal subsets. In each iteration, one subset combined with an equal number of randomly sampled unknown triplets served as the test set, while the remaining four subsets along with other unknown samples constituted the training set. Method performance was evaluated using the area under the precision-recall curve (AUPR), the area under the receiver operating characteristic curve (AUC), and F1-score. This evaluation setting assesses the capability of DHGCMDB in discovering novel miRNA-disease associations from the vast space of unexplored pairs.
- **CV_{type}**: All miRNA-disease pairs with at least one known association type were identified and randomly divided into 5 subsets. In each fold, four subsets were used for training while the remaining subset was held out for testing. For each test pair, DHGCMDB generated prediction scores across all four association types, with the highest-scoring type selected as the final prediction. Top-1 precision, Top-1 recall, and Top-1 F1-score were calculated to evaluate discriminative performance. This evaluation setting focuses on the discriminative ability of DHGCMDB in identifying specific biological mechanisms underlying known associations.

These two complementary evaluation settings assess performance from both association discovery and mechanism identification perspectives, consistent with dual objectives emphasized in recent miRNA-disease association type prediction studies.

All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 4060 Ti GPU (16 GB VRAM) and 32 GB RAM, implemented using PyTorch 2.5.1 with CUDA 12.1. The total number of trainable parameters in DHGCMDB is approximately 2.55M. Under the 5-fold cross-validation setting on HMDD v2.0, the average training time per fold was approximately 3.16 minutes, with the total cross-validation completing in about 15.80 minutes. The peak GPU memory usage was 3.68 GB across all folds, indicating that DHGCMDB maintains moderate

computational overhead and can be readily deployed on consumer-grade hardware without requiring high-end computing resources.

3.3 Hyperparameter analysis

The sensitivity of DHGCMDA performance to hyperparameters is investigated based on the HMDD v2.0 dataset under the CV_{type} evaluation setting. Key hyperparameters, including the temperature parameter t (Eq. 6), the cross-modality contrastive loss weight λ_2 (Eq. 32), and the class weights w_k (Eq. 34), are analyzed in detail.

Grid search was performed over t and λ_2 with w_k fixed to uniform weighting. Specifically, the search range of λ_2 was set to $\{0.1, 0.3, 0.5\}$ and that of t to $\{0.5, 1.0, 3.0\}$. As shown in Fig. 2(a), all Top-1 metrics of DHGCMDA reached maximum values when t and λ_2 were set to 0.5 and 0.3, respectively. Furthermore, larger values of λ_2 led to performance degradation. This phenomenon may indicate that excessive emphasis on cross-modality alignment could overshadow discriminative information captured within individual modalities. The temperature parameter exhibited remarkable stability across all tested values, demonstrating that the dual-view hypergraph construction and attention-guided fusion mechanisms provide sufficient discriminative capacity, thus reducing sensitivity of DHGCMDA to temperature variations.

Using the optimal hyperparameters of t and λ_2 , three class weighting strategies were further compared: inverse proportional weighting, uniform weighting, and square-root inverse weighting. As shown in Fig. 2(b), the inverse proportional strategy exhibited superior performance over the other two strategies, suggesting that appropriate emphasis on minority association types facilitates the ability of DHGCMDA to capture diverse biological mechanisms underlying miRNA-disease associations.

Regarding the additional trade-off hyperparameters λ_1 and λ_3 in the unified training objective (Eq. 32), both were set to 1.0 in this study. To prevent the introduction of excessive tunable parameters and mitigate the risk of overfitting on relatively small

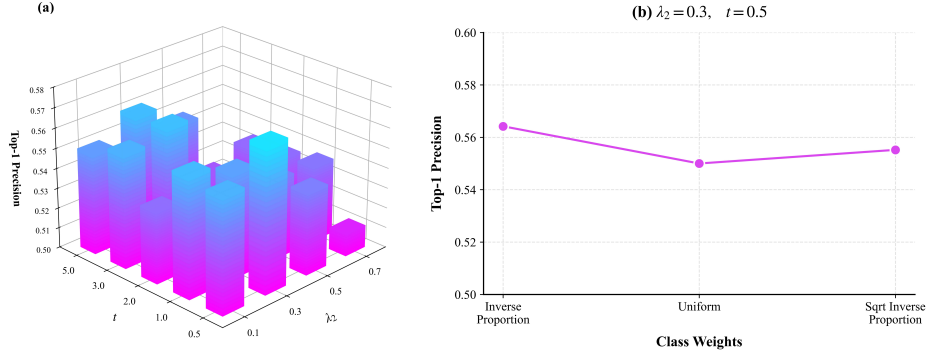


Fig. 2 Hyperparameter analysis of DHGCMMA on the HMDD v2.0 dataset: (a) sensitivity to t and λ_2 ; (b) comparison of class weighting strategies.

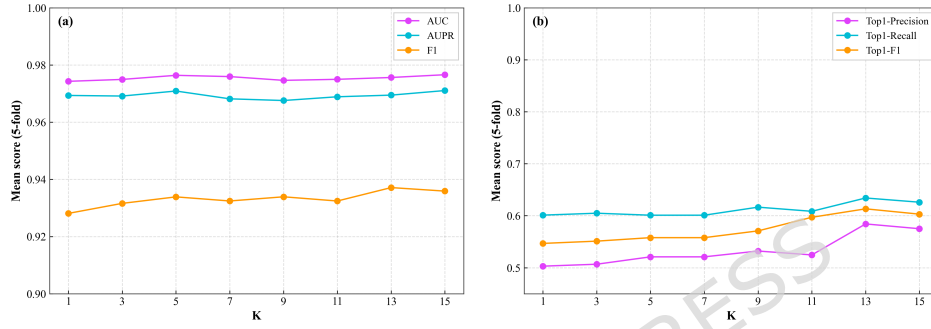


Fig. 3 Hyperparameter analysis of DHGCMMA on the HMDD v2.0 dataset: (a) binary classification metrics vs K ; (b) Top-1 metrics vs K .

biological datasets, we treated the intra-modality contrastive loss and the similarity reconstruction loss as equally weighted auxiliary objectives alongside the primary classification loss. Consequently, we focused our sensitivity analysis on λ_2 , which directly governs the inter-modality alignment and exerts a more decisive influence on the final association type prediction performance. Furthermore, we evaluated the sensitivity of the model to the neighborhood size K used for hypergraph construction. As illustrated in Fig. 3, binary classification metrics (AUC, AUPR, and F1) remain stable across $K \in \{1, 3, 5, 7, 9, 11, 13, 15\}$, whereas Top-1 metrics exhibit an overall upward trend and reach their maximum at $K = 13$, followed by a slight decline at $K = 15$. Consequently, $K = 13$ is adopted for all subsequent experiments.

3.4 Comparison with other methods

To evaluate the performance of DHGCMDA, comparative experiments were conducted against six state-of-the-art methods for miRNA-disease association type prediction, namely TDRC [22], SPLDHyperAWNTF [24], TFLP [25], NMCMDA [28], MRFGMDA [33], and KBLTDARD [34]. The experiments were carried out on the HMDD v3.2 and HMDD v2.0 datasets under the CV_{type} and $CV_{triplet}$ evaluation settings. The comparative results are presented in Table 3 and Table 4, respectively.

As shown in Table 3, DHGCMDA significantly outperforms all methods on HMDD v3.2 and achieves competitive performance on HMDD v2.0 under the CV_{type} . Specifically, on HMDD v3.2, DHGCMDA delivers a Top-1 precision of 0.7915, recall of 0.9421 and F1-score of 0.8600, with respective gains of 25.61%, 45.36% and 35.75% over the suboptimal values of each metric derived from different comparison methods. On HMDD v2.0, DHGCMDA achieves the highest Top-1 recall among all methods. While its precision lags behind that of several comparison methods, DHGCMDA still maintains a top-tier F1-score. These remarkable gains on HMDD v3.2 stem from dual-view hypergraph and contrastive learning modules that effectively capture high-order biological relationships, advantages single-graph or linear methods lack when handling complex data distributions. The moderate precision on HMDD v2.0 is a deliberate trade-off via class weighting, which prioritizes true association recall to ensure optimal F1-score balance for biomedical prediction tasks.

As shown in Table 4, DHGCMDA achieves dominant performance across both datasets under the $CV_{triplet}$. Specifically, on HMDD v2.0, DHGCMDA ranks first in AUPR, AUC and F1-score, with respective gains of 2.18%, 2.21% and 3.93% over KBLTDARD, the second-best method. On HMDD v3.2, DHGCMDA still outperforms all comparison methods in AUPR and F1-score, a result that demonstrates stable performance across different datasets. AUPR and AUC act as critical metrics for

Table 3 Performance comparison under CV_{type} on the HMDD v3.2 and HMDD v2.0.

Method	HMDD v3.2			HMDD v2.0		
	Top-1 Precision	Top-1 Recall	Top-1 F1-score	Top-1 Precision	Top-1 Recall	Top-1 F1-score
MRFGMDA	0.6229	0.6481	0.6335	0.5860	0.6152	0.5979
NMCMMDA	0.6199	0.4609	0.5287	0.6061	0.5409	0.5716
SPLDHyperAWNTF	0.6219	0.4624	0.5304	0.5050	0.4506	0.4762
TFLP	0.4812	0.3553	0.4076	0.6358	0.5673	0.5996
KBLTDARD	0.6301	0.4685	0.5147	0.6122	0.5464	0.5683
TDRC	0.4926	0.3671	0.4207	0.5090	0.4542	0.4801
DHGCMDA	0.7915	0.9421	0.8600	0.5842	0.6341	0.5970

Table 4 Performance comparison under $CV_{triplet}$ on the HMDD v3.2 and HMDD v2.0.

Method	HMDD v3.2			HMDD v2.0		
	AUPR	AUC	F1-score	AUPR	AUC	F1-score
MRFGMDA	0.8845	0.8845	0.8694	0.9004	0.8893	0.8421
NMCMMDA	0.8885	0.8681	0.8352	0.9280	0.9329	0.8765
SPLDHyperAWNTF	0.9126	0.9102	0.8395	0.9253	0.9091	0.8584
TFLP	0.8217	0.7229	0.7831	0.9461	0.9395	0.8843
KBLTDARD	0.9247	0.9263	0.8541	0.9530	0.9460	0.8927
TDRC	0.9059	0.8962	0.8309	0.9085	0.8973	0.8578
DHGCMDA	0.9271	0.9181	0.8674	0.9738	0.9669	0.9278

evaluating method performance on imbalanced biological data, so the superior performance of DHGCMDA in these two metrics indicates robust ability to identify minority miRNA-disease association types. These results verify that the attention-guided fusion mechanism in DHGCMDA can effectively integrate multi-view feature representations, thereby boosting the accuracy of miRNA-disease triple link prediction.

3.5 Ablation Study

To further validate the effectiveness of each component in DHGCMDA, ablation experiments were conducted on the HMDD v2.0 and HMDD v3.2 datasets under the CV_{type} and $CV_{triplet}$ evaluation settings. Five variant models were constructed by removing or replacing core components to dissect their individual contributions to the overall performance:

- **w/o CL**: Both intra-modality and cross-modality contrastive learning modules were removed.
- **w/o HGNCN**: The HGNCN was replaced with a standard GCN.
- **w/o AVF**: The attention-guided adaptive view fusion was substituted with simple feature concatenation.
- **w/o HGT**: The HGT was eliminated, and fused embeddings were directly adopted for prediction.
- **w/o DV**: Only a single-view hypergraph was employed instead of the dual-view construction.

The prediction performance of all variants is presented in Fig. 4. Among these variants, w/o DV exhibits the most dramatic performance degradation, which highlights that integrating multi-source biological similarity data is indispensable for comprehensively capturing the characteristics of miRNAs and diseases. This result can be attributed to the strong complementarity between the two biologically heterogeneous similarity views: the sequence-based view captures structural and evolutionary properties of miRNAs, while the functional view encodes their regulatory roles in biological pathways. Similarly, the gene-based and semantic views of diseases provide orthogonal characterizations from molecular and phenotypic perspectives, respectively. When only a single view is retained, the model loses access to these complementary signals, leading to embeddings that are biased toward one aspect of biological similarity and thus insufficient for distinguishing among diverse association types. The dual-view design, combined with contrastive learning, enables the model to distill view-invariant features that are robust to view-specific noise, which is particularly critical for fine-grained type prediction where subtle differences in biological mechanisms must be captured. Likewise, w/o HGT leads to substantial performance declines, indicating that the type-aware message passing mechanism effectively models the complex interactions between miRNAs and diseases. The w/o HGNCN variant also performs markedly

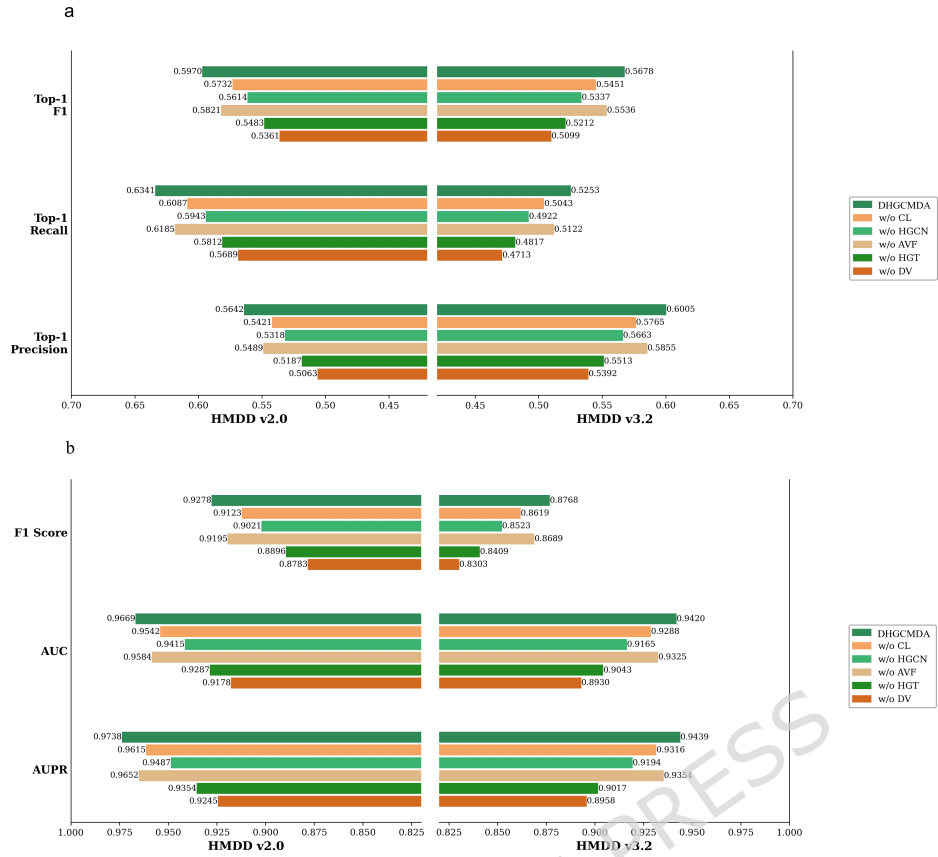


Fig. 4 Ablation results under CV_{type} and $CV_{triplet}$ on the HMDD v2.0 and HMDD v3.2.

worse than DHGCMDA, demonstrating that HGCN surpasses standard GCN in capturing high-order relationships within biological networks. Additionally, the decreased performance of w/o CL confirms that contrastive learning enhances the discriminative power of the learned feature representations. Collectively, these results demonstrate that all five components play critical roles in ensuring the optimal performance of DHGCMDA, and their synergistic effects are essential for achieving superior predictive performance in miRNA-disease association type prediction.

3.6 Case Study

To further evaluate the capability of DHGCMDBA in predicting novel miRNA-disease association types, case studies were conducted on breast neoplasms and hepatocellular carcinoma. DHGCMDBA, trained on the HMDD v2.0 dataset, was employed to predict miRNA-disease-type triplets. For each disease, miRNA-type pairs were ranked by their prediction scores, with the top 15 ones selected as candidates. Supporting evidence for these candidates was retrieved and verified via extensive literature searches in the PubMed database, and validation results are summarized in Table 5 and Table 6, corresponding to breast neoplasms and hepatocellular carcinoma, respectively.

For breast neoplasms, 13 out of 15 predicted miRNA-disease association types (86.7% validation rate) were successfully confirmed by published studies. For instance, hsa-mir-148a, whose epigenetics type association was validated by evidence that it regulates estrogen receptor- α expression through DNMT1-mediated DNA methylation in breast cancer cells (PMID: 24257477); hsa-mir-195, identified as a reliable circulation biomarker with significantly elevated levels in the plasma of breast cancer patients compared with healthy controls (PMID: 25103018); hsa-mir-125b, whose genetics type association was established via the discovery of a risk variant in the BMPRI1B 3'UTR binding site that disrupts miRNA binding and increases cancer susceptibility (PMID: 19738052); and hsa-mir-1290, confirmed as a target type miRNA that directly targets NAT1 and exhibits subtype-specific downregulation in ER-positive breast tumors (PMID: 23183268).

For hepatocellular carcinoma, 12 out of 15 predicted miRNA-disease association types (80% validation rate) were successfully validated by published studies. Typical examples include, hsa-mir-196a, whose epigenetics type association was confirmed by evidence showing that it modulates tumor suppressor gene expression through histone modification in hepatocellular carcinoma cells (PMID: 24377574); hsa-mir-217, a target type miRNA that directly targets the oncogene STAT3 and inhibits cancer

cell proliferation (PMID: 23471579); hsa-mir-26b, whose genetics type association was established via the identification of a functional polymorphism in its promoter region linked to increased hepatocellular carcinoma risk (PMID: 26891666).

Notably, two breast neoplasms associated and three hepatocellular carcinoma associated miRNAs were labeled as “Unconfirmed”. This designation does not indicate flaws in the model predictions; instead, it reflects the absence of direct evidence supporting these specific functional types in published literature. In fact, all these miRNAs have been confirmed to be associated with their corresponding cancers, except that their actual association types are inconsistent with the model predictions. This phenomenon suggests that these unconfirmed associations may represent potential novel miRNA-disease-type triplets, which need further targeted experimental validation. Specifically, these predictions could constitute biologically plausible novel discoveries that merit prioritized wet-lab validation, such as qRT-PCR verification of circulating miRNA levels or functional assays to confirm the predicted mechanism types. Alternatively, a proportion of these cases may reflect model uncertainty in distinguishing closely related association types, particularly when the biological boundaries between mechanisms are ambiguous. Nevertheless, the small proportion of unconfirmed cases (5 out of 30 total predictions, 16.7%) suggests that the overall prediction reliability of DHGCMDA remains high.

To explore the biological underpinnings of the predicted miRNA-disease association types, enrichment analysis was performed on target genes of breast neoplasms associated miRNAs using miRTarBase [4] and Metascape [35], with results summarized in Fig. 5. Specifically, Fig. 5(a) shows DisGeNET disease association analysis, which revealed marked enrichment in breast carcinoma stage IV, and thus directly confirmed the functional relevance of predicted miRNAs to breast cancer progression. Fig. 5(b) displays protein-protein interaction network analysis combined with MCODE clustering, which identified functionally coherent modules encompassing growth factor

signaling molecules (EGFR, ERBB2, ERBB3) and SMAD pathway members (SMAD1, SMAD4, SMAD5). Importantly, ERBB2 is highly expressed in approximately 15–20% of breast cancers [36] and represents a well established clinical therapeutic target, whereas the TGF β /SMAD pathway acts as a key regulator of breast cancer metastasis [37]. Fig. 5(c) presents GO and KEGG pathway enrichment analysis, in which significant enrichment was observed in cell cycle related pathways (WP179 Cell cycle), consistent with the hallmark of uncontrolled cancer cell proliferation [38], as well as TGF β /SMAD and EGFR/ERBB signaling pathways that are closely linked to breast cancer metastasis and clinical therapeutic relevance. Collectively, these results demonstrate that DHGCMDA can reliably predict biologically meaningful miRNA disease association types with specific functional mechanisms. From a biological perspective, the identified pathway modules provide actionable insights into the regulatory roles of predicted miRNAs. The enrichment of ERBB2-related signaling modules suggests that predicted target-type miRNAs (e.g., hsa-mir-125b and hsa-mir-130a) may exert their regulatory effects by modulating receptor tyrosine kinase cascades central to breast cancer cell proliferation and survival. Similarly, the TGF β /SMAD pathway enrichment aligns with predicted epigenetics-type associations (e.g., hsa-mir-148a and hsa-mir-335), as this pathway is known to be regulated by epigenetic modifications including DNA methylation and histone acetylation. These pathway-level findings not only corroborate the predicted association types but also suggest potential mechanistic hypotheses that could guide future experimental validation.

4 Conclusion

MiRNA-disease association type prediction provides a critical computational tool for dissecting the molecular pathogenic mechanisms underlying human diseases driven by aberrant miRNA expression. In this study, DHGCMDA, a dual-view heterogeneous

Table 5 Top 15 breast neoplasms related miRNAs.

miRNA	Association Type	PMID
hsa-mir-148a	epigenetics	24257477
hsa-mir-1290	target	23183268
hsa-mir-449b	genetics	unconfirmed
hsa-mir-195	circulation	25103018
hsa-mir-130a	target	25755726
hsa-mir-632	target	22710984
hsa-mir-148b	circulation	24194846
hsa-mir-335	epigenetics	21289068
hsa-mir-4521	target	24517586
hsa-mir-30c	target	24519092
hsa-mir-593	epigenetics	unconfirmed
hsa-mir-125b	genetics	19738052
hsa-mir-488	genetics	16754881
hsa-mir-1323	circulation	22242178
hsa-mir-24	genetics	24966325

Table 6 Top 15 hepatocellular carcinoma related miRNAs.

miRNA	Association Type	PMID
hsa-mir-196a	epigenetics	24377574
hsa-mir-217	target	23471579
hsa-mir-429	epigenetics	24572141
hsa-mir-26b	genetics	26891666
hsa-mir-411	target	25776495
hsa-mir-1469	circulation	unconfirmed
hsa-mir-30d	genetics	27333771
hsa-mir-204	target	24833879
hsa-mir-487b	circulation	unconfirmed
hsa-mir-766	target	30130435
hsa-mir-153	genetics	25714700
hsa-mir-657	target	23175432
hsa-mir-19b	genetics	unconfirmed
hsa-mir-133a	target	26156803
hsa-let-7d	target	20347499

graph contrastive learning framework that integrates dual-view hypergraph convolution and heterogeneous graph Transformer, is proposed for miRNA-disease association type prediction. Specifically, dual-view hypergraphs are first constructed based on heterogeneous similarity data to avoid excessive reliance on association-derived similarity metrics. The HGCN is then employed to capture high-order topological relationships

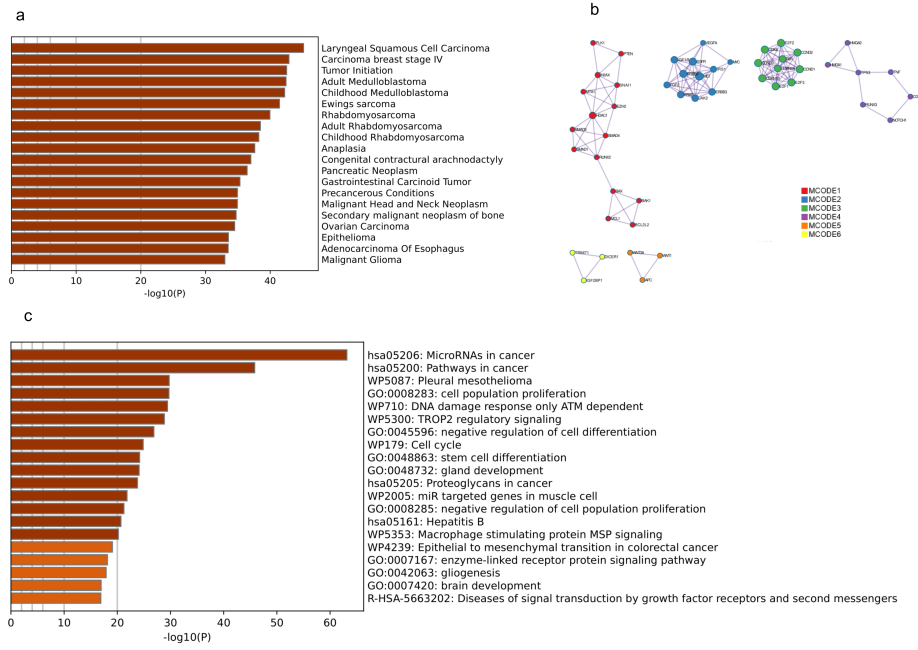


Fig. 5 Enrichment analysis of target genes from predicted breast neoplasms associated miRNAs.

between miRNAs and diseases, with its convolution cooperatively integrated with contrastive learning, intra-modality for cross-view consistency and cross-modality for embedding space alignment, to enhance feature representation quality. Finally, an attention-guided adaptive view fusion strategy dynamically weights and integrates distinct view representations, and type-aware message passing via HGT simultaneously enables prediction of association presence and functional types. The experimental results demonstrate that DHGCMDDA achieves superior prediction performance compared to six state-of-the-art models under CV_{type} and $CV_{triplet}$ on both HMDD v2.0 and HMDD v3.2 datasets. In addition, the ablation experiments further demonstrate the effectiveness of different components, particularly showing that dual-view hypergraph construction and HGT are critical for optimal performance. Moreover, the results of case studies show that DHGCMDDA can accurately predict miRNA-disease association types and the predicted associations have biological significance, which further validate the effectiveness and feasibility of DHGCMDDA.

Despite the promising results, DHGCMDBA has several limitations that warrant discussion. First, the current framework relies on the availability of multiple similarity measures for both miRNAs and diseases; for novel miRNAs or rare diseases where such data are extremely sparse, the dual-view construction may degrade to a near-single-view setting, potentially limiting performance gains. Second, the computational cost of DHGCMDBA scales with the number of nodes and hyperedges, and applying it to significantly larger biological networks may require architectural optimizations such as mini-batch hypergraph sampling or more efficient attention mechanisms. Third, while DHGCMDBA learns expressive node embeddings, the interpretability of these embeddings remains limited; future efforts could incorporate attention visualization or gradient-based attribution methods to provide mechanistic insights into predicted association types. Fourth, the dynamic hypergraph update strategy currently employs a fixed threshold θ and update interval, which may not be optimal across all training stages; adaptive thresholding strategies could further improve structural refinement.

Since DHGCMDBA is a general framework for multi-relational link prediction in heterogeneous biological networks, it can be readily extended to other tasks such as drug-drug interaction type prediction and lncRNA-disease association type prediction. In future work, efforts will be directed toward addressing the limitations identified above: integrating additional biomedical information sources such as miRNA-gene interactions to mitigate data sparsity, adopting positive-unlabeled learning strategies to better handle uncertainty in negative samples, and exploring adaptive graph updating mechanisms and model interpretability techniques to enhance both structural refinement and biological insight.

Acknowledgements. The authors would like to thank the reviewers for their insightful comments and suggestions that helped improve the quality of this manuscript. This research was supported in part by computational resources provided by the School of Computer Science, Qufu Normal University.

Declarations

Funding. This work was supported by the National Natural Science Foundation of China (62472250, 62473179).

Conflict of interest. The authors declare no conflicts of interest.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Availability of data and materials. The dataset supporting the conclusions of this article is available in the HMDD repository, <https://www.cuilab.cn/hmdd>. The implementation code of DHGCMDBA is publicly available at <https://github.com/CDMBlab/DHGCMDBA>.

Authors' contributions. Yan Sun: Methodology and Writing - original draft. Fanyu Zhang: Formal analysis and Writing - review and editing. Shijia Yan: Visualization. Xiaotong Kong: Investigation. Hanxiang Wang: Validation. Junliang Shang: Funding acquisition and Writing - review and editing. Jin-Xing Liu: Supervision and Funding acquisition.

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